Better Spotify / Sample Browser

Granular Audio Passage Discovery for Independent Music Producers

Problem-Based Learning Journey: Evolving from a Passive Playback Concept into an Active Creator Workspace

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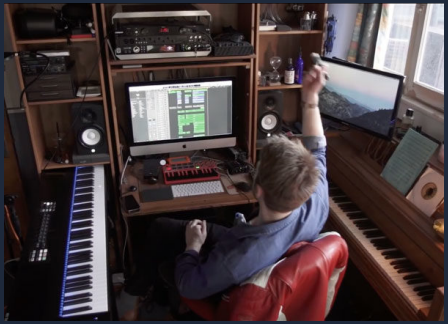
PROJECT SUPERVISION

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The Environment and Reality of Audio Sampling



Production Context: Bedroom workstation with DAW, keyboard, and monitoring.

CORE CONCEPT

Sampling: Reusing a portion of an existing recording in a new composition.

Producers hunt for localized **2–8 bar passages** (4–30s), not whole songs.

17%

Billboard Hot 100

Hits rely on samples (Tracklib)

250M+

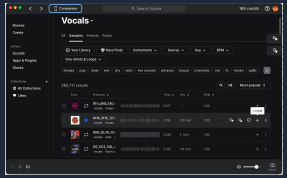
Streaming Catalog

Tracks indexed mainly at track level

Three Existing Paradigms vs. Passage Discovery

Splice

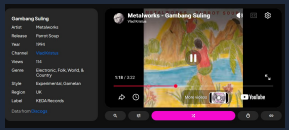
Ready-Made Samples



Catalog of pre-cut loops & one-shots; prepared library workflow.

Samplette.io

Discover Full Recordings



Randomized YouTube digging via Discogs; whole-track search only.

Tracklib

Licensed Catalog + Tools



Pre-cleared songs with stem isolation and bar-looping tools.

Sample Browser: Discover and compare passages across recordings.

The Core Problem & Strategic Workflow Pivot

Producers lack an efficient way to find specific passages inside full recordings.

Streaming catalogs index music as 4-minute monoliths; producers search by sound timbre for 2–8 bar passages.



Current Scrubbing Loop:

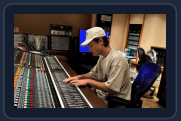
Full Track ⟶ Manual Scrub ⟶ Repeat / Flow Lost

Proposed Passage Workflow:

Describe Sound ⟶ Find Passage ⟶ Jump to Cue

PBL Strategic Pivot: Consumer streaming shell ⟶ Spotify API locks (audio endpoints deprecated) ⟶ **Creator-focused passage search**

Two Illustrative Creator Personas (Validation Planned)



Alex Miller

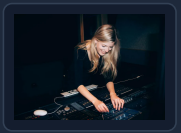
Aspiring Producer | 22

- **Role:** Bedroom producer (FL Studio)
- **Time:** 8–12 hrs/week creative time
- **Friction:** Scrubbing full-length tracks

Scenario:

Found vocal

→ **needs matching guitar**



Annie Lee

Hobbyist Producer | 20

- **Role:** Student sound designer
- **Time:** Exploratory / sporadic
- **Friction:** Generic sample pack fatigue

Scenario:

Found interesting sound

→ **cannot find similar passages**

Target Market Distribution (Tracklib Survey, $N \approx 1{,}000$): **58%** Aspiring (Alex) | **25%** Hobbyists (Annie) | **10%** Professional Producers ($\approx$ 7% Beginners)

05. THE SOLUTION

Find the passage, not just the song.



Concept mockup (illustrative; match scores are not from a working algorithm)

1. Acoustic Filters

Instrument, texture, timbre

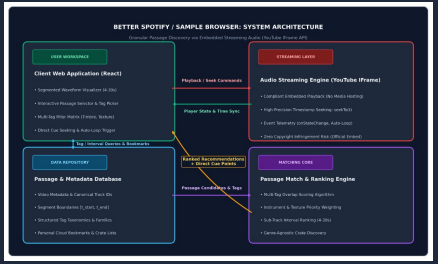
2. Passage Results

Timestamped section candidates

3. Direct Cue + Loop

Instant playhead jump (`seekTo`)

Interaction Pipeline & Technical Scope



System Architecture: Client UI, YouTube IFrame player, and PostgreSQL metadata backend.

Interaction Pipeline

Web UI ⟶ **Metadata** ⟶ **Timestamp** ⟶ **YouTube Player**

Implementation Status

Current: Curated acoustic tagging and timestamp metadata.

Planned (M2): Automated MIR indexing (Essentia classifiers).

*Note: MIR indexing is planned for evaluation, not implemented.

Technical Caveat: Compliant audio ingestion for server-side MIR remains unresolved. No original audio hosting; user clears sampling rights.

Competitive Positioning & Strategic SWOT Overview

Competitor Landscape

- **Splice:** Curated loop packs (isolated samples)
- **Samplette:** Whole-track YouTube discovery
- **Tracklib:** Pre-cleared catalog + looping tools
- **Sononym:** Local desktop similarity search
- **Cyanite:** Enterprise B2B sync intelligence

Proposed focus: a creator workflow combining acoustic filters, passage-level search across recordings, and direct timestamp auditioning.

Strategic SWOT Overview

Strengths (S)

- Sub-track granularity
- Direct timestamp cueing
- No original audio hosting

Weaknesses (W)

- Cold-start catalog
- Stream dependency
- Unresolved MIR ingest

Opportunities (O)

- Sample pack fatigue
- Future DAW integration
- Label partnerships

Threats (T)

- Streaming API shifts
- Incumbent feature creep
- Clearance friction

*Full comparative matrix and complete SWOT analysis preserved in **Appendix (Slides 10–11)** for committee Q&A.

Engineering Roadmap & Validation Hypotheses

1. Validate

10–15 Producer Interviews

Direct interviews on workflow friction and digging habits.

2. Prototype (M2)

Interactive UI & Architecture

Functional wireframing and UML structural models.

3. Evaluate (Exam)

Passage Utility Assessment

Empirical evaluation of passage retrieval utility.

Research Hypotheses (To Investigate):

[Scrubbing Pain (H1)] **[Tool Gap (H2)]** **[Time Target (H3)]**

Field validation planned across $N = 10–15$ creator interviews

Potential Business Model (To Be Validated):

Free Community Tier | **Pro Creator Tier**

Pro Tier (hypothesis): Timbral MIR similarity, DAW marker exports (MIDI/cue), unlimited crates

*Monetization viability remains an unvalidated hypothesis.

Takeaway: Sample Browser explores a missing link between finding a recording and finding the exact passage worth using.

APPENDIX · Q&A REFERENCE

Strategic SWOT Analysis (Report Section 2.4, Table 3)

STRENGTHS (S)

- **Sub-Track Granularity:** Indexes at 2–16 bar interval level, designed to reduce manual scrubbing.
- **Explainable Discovery:** Transparent acoustic descriptors rather than opaque black-box recommendations.
- **No Original Audio Hosting:** Uses compliant YouTube IFrame streaming without media storage overhead.
- **Contextual Playback:** Audition passages in original arrangements.

WEAKNESSES (W)

- **Cold-Start Catalog Challenge:** Platform utility in early stages relies heavily on pre-tagged passage volume.
- **Third-Party Dependency:** Vulnerable to YouTube video takedowns, regional geo-blocks, or API policy changes.
- **No Direct Audio Export:** Users must acquire source audio externally under API terms.

OPPORTUNITIES (O)

- **Sample Pack Fatigue:** Producer backlash against overused commercial sample libraries (Splice).
- **DAW Plugin Integration:** Potential to port interface into embedded VST/AU inside Ableton or FL Studio.
- **Catalog Rights Monetization:** Partnership pathways with archival labels to monetize dormant back-catalogs.

THREATS (T)

- **API Policy Lockdowns:** Risk of streaming providers restricting embedded player functionality.
- **Incumbent Expansion:** Established tools (Tracklib, Samplette) expanding into segment-level tagging.
- **Copyright Uncertainties:** Legal clearance requirements for archival recordings in commercial releases.

APPENDIX · Q&A REFERENCE

Formal Problem Statement & Evaluation Matrix

VERBATIM FORMAL PROBLEM STATEMENT (REPORT SEC. 2.1)

"Independent music producers lack an efficient way to discover and locate specific musical passages within full-length catalog recordings based on acoustic references, because contemporary streaming search interfaces index music exclusively at the whole-track metadata level, forcing producers into tedious, manual scrubbing that disrupts the creative workflow."

Comparative Tool Matrix (Report Table 2)

Platform	Index Granularity	Query Paradigm	Catalog Type	Workflow Role
Splice	Isolated samples	Text / Metadata	Curated proprietary	Ready-made sound packs
Samplette	Full-track	Random / Discogs	Public (YouTube)	Full-record digging
Tracklib	Track + Bar loops	Metadata / Mood	Pre-cleared master	Licensing marketplace
Sononym	Local sample files	Audio similarity	Local user disk	Sample library organizer
Cyanite.ai	Track & Segment	Audio similarity	Enterprise B2B	Sync intelligence API
Sample Browser	**Passage (4–30s)**	**Acoustic descriptors**	**Public (YouTube)**	**Cross-catalog discovery**